

Case no. 1

A 50-year-old woman presented with right upper abdominal pain, fever and nausea for the past week.

The patient states that the pain is radiating to the back and right shoulder blade.

O/E:

Height is 160 cm, weight is 90 kg, BMI 33.

Febrile, no jaundice.

Marked tenderness of the right upper quadrant, liver span is normal.

Case no. 1

LAB findings include:

Fasting glucose = 190 mg/dL;
Creatinine = 0.8 g/dL;
Ferritin = 250 ng/mL;
Total protein = 7 g/dL; and albumin = 4.3 g/dL.
TSB, AST, ALT, and ALP are normal.

Total protein.	= 6.3 - 7.9 g/dL.
Albumin.	= 3.5 - 5.0 g/dL.
Bilirubin.	= 0.1 - 1.2 mg/dL.
ALT.	= 7 - 55 U/L.
AST.	= 8 - 48 U/L.
ALP.	= 40 - 129 U/L.
Ferritin	= 30 - 400 ng/ mL
PT.	= 9.4 - 12.5 sec.
PTT.	= 25 - 35

Hemoglobin = 13.5 g/dL, hematocrit = 42%, MCV = 94 μm^3 .
Platelet count = 250,000/mm³.
WBC count = 13500/mm³.
PT and PTT are normal.

Case no. 1/ Questions

Abdominal US shows stones within the lumen of the gallbladder, and the gallbladder wall appears thickened.

Intrahepatic and extrahepatic bile ducts appear normal.

What is the diagnosis?

What is the likely type of the stone?

What is the pathogenesis of the disease?



Case no. 1

Which of the following factor(s) do play a role in development of her disease? How?

- A. Biliary hypersecretion of cholesterol.
- B. Chronic hemolysis.
- C. Decrease ceruloplasmin level.
- D. Hepatocyte infection by HBV.
- E. High blood LDL level.
- F. Increased intestinal absorption of iron.
- G. Increased estrogen level.
- H. Ingestion of foods rich in fat.
- I. Prolonged fasting.
- J. Rapid weight loss.
- K. T2DM.

Case no. 1/ Questions

- The patient states that “the pain is radiating to the back and right shoulder blade” .
- What is the possible mechanism?

Case no. 1/ Questions

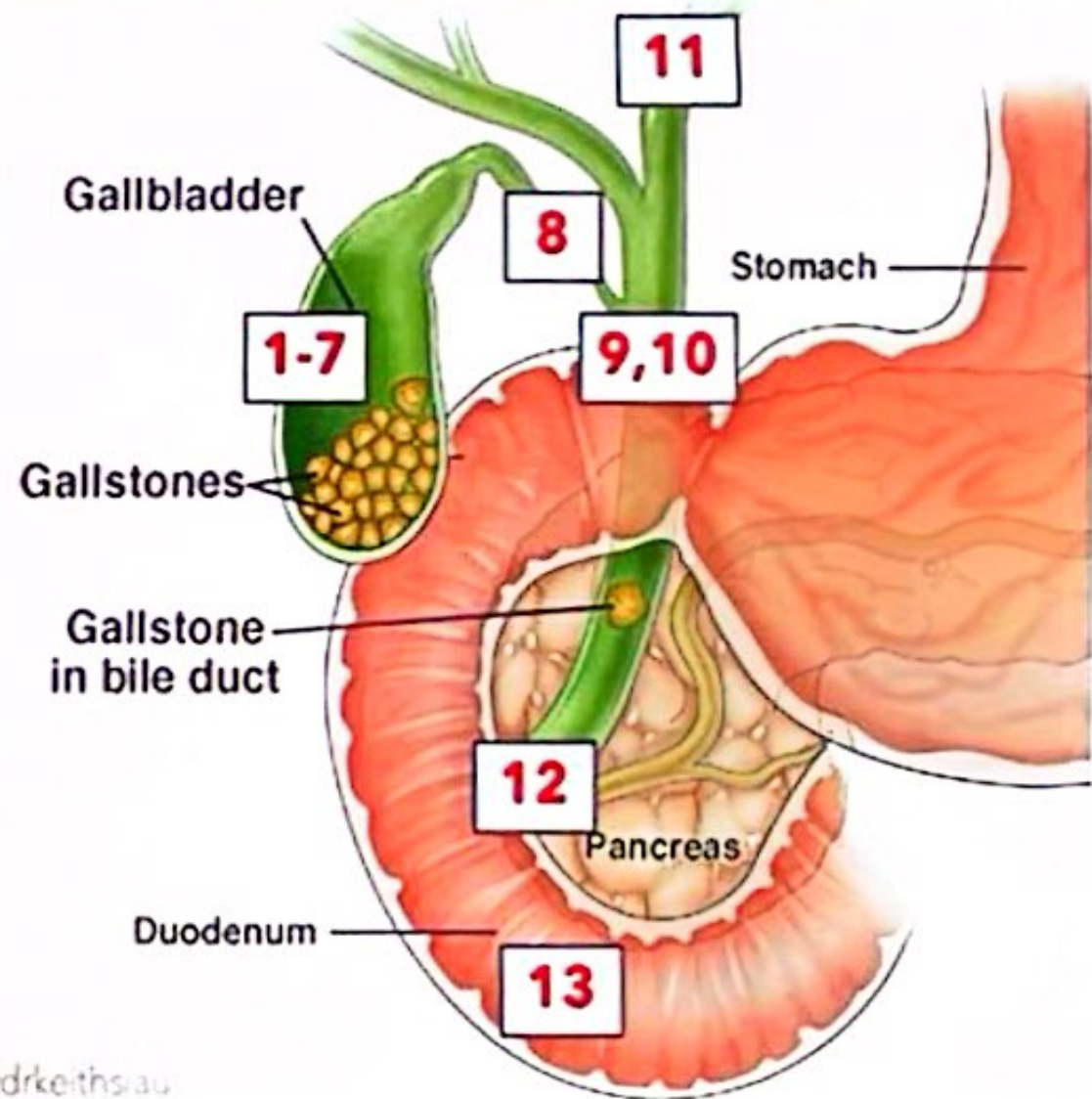
- What possible complications are expected to occur if the stones remain in the gall bladder?
1. Biliary colic.
 2. Cholecystitis.
 3. Mucocele.
 4. Empyema.
 5. Perforation.
 6. Fistula.
 7. Gallbladder carcinoma.

Case no. 1/ Questions

- What possible complications are expected to occur if the stones move out the gall bladder?

Outcomes and complications of GS

1. Biliary colic.
2. Cholecystitis.
3. Mucocele.
4. Empyema.
5. Fistula.
6. Perforation.
7. Gallbladder carcinoma.
8. Common hepatic duct obstruction.
9. Choledocholithiasis.
10. Cholangitis.
11. Intrahepatic stone.
12. Pancreatitis.
13. Gall stone ileus.



Case no. 1/ Questions

- What is the explanation for pain, fever and leukocytosis?

Case no 2/ Presentation:

- A 43-year-old man has experienced progressive fatigue, pruritus, Mild right upper abdominal pain, icterus and passing dark urine for several months.
- PMHx: Diagnosed with ulcerative colitis at age 27 yrs, on remission.
- O/E:
Jaundice.
Abdomen: Mild tenderness in the right upper quadrant.
No organomegally, no masses.

Case no 2/ Presentation

- Liver function tests:

- ALT: 73 U/L, AST: 65 U/L
- ALP: 290 U/L
- Total Billirubin: 2.6 mg/dL, conj bilirubin: 2 mg/dL
- Total protein: 5.9 g/ dL
- Ferritin= 270 ng/mL

- Viral hepatitis serologies: Negative.

- Serology for autoAB:

- Antimitochondrial antibodies (AMA) : Negative.
- Antinuclear antibodies (ANA) : Negative.
- Anti-smooth muscle antibodies (ASMA) : Negative.
- Anti-neutrophil cytoplasm antibodies (ANCA) : Positive

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ALT.	= 7 - 55 U/L
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Case no 2/ Presentation:

- Imaging:

Liver shows widespread obliteration of intrahepatic bile ducts and a beaded appearance in the remaining extrahepatic ducts.

Case no 2/ Questions:

- What is the most likely diagnosis?
- What other diagnostic tests do we have to confirm the diagnosis?

Case no 2/ Questions:

- Which of the following histologic features is most likely to be present in this patient's liver biopsy?
 - A. Non suppurative granulomatous reaction.
 - B. Concentric "onion-skin" bile duct fibrosis.
 - C. Copper deposition in hepatocytes
 - D. Deposition of brownish pigments stains with pearl stain within macrophages.
 - E. Peri-portal PAS-positive globules

Case no 2/ Questions:

- What is the type of jaundice and what does it indicate?
- What is the cause of pruritis?
- What explains the dark colored urine?

Case no 2/ Questions:

- What are long term complication risks does this disease have?